1 Appendix

1.1 Theoretical Guarantees

First we present lemmas, which will support our theorems. We note that the this analysis goes more or less along the same lines of [1].

Let us define the event $\mathcal{F}_p(t) = \left\{ \exists i \in [N],\ j \in [K] : |\hat{\mu}_j^{(i)} - \mu_j^{(i)}| > \sqrt{\frac{2\log t}{T_j^{(i)}}} \right\}$, a player's bad event that some preference of the arms are not estimated by the players correctly at time t. We present the following Lemma 5.1, Lemma 5.4 ,and Lemma 5.5 of [1]

Lemma 1.1.

$$\mathbb{E}\left[\sum_{t=1}^{T} 1\{\mathcal{F}_p(t)\}\right] \leq \frac{NK\pi^2}{3} \tag{1}$$

Lemma 1.2. *Conditional on $\mathcal{F}_p(t)^{\complement}$, $\mathrm{UCB}_j^{(i)}(t) < \mathrm{LCB}_{j'}^{(i)}(t)$ implies $\mu_j^{(i)} < \mu_{j'}^{(i)}$.*

Lemma 1.3. *In round t, let $T^{(i)}(t) = \min_{j\in[K]} T_j^{(i)}(t)$ and $\bar{T}^{(i)} = 32\log T/(\Delta^{(i)})^2$ where $\Delta^{(i)} = \min_{j\neq j'} \Delta_{j,j'}^{(i)} = \min_{j\neq j'} |\mu_j^{(i)} - \mu_{j'}^{(i)}|$. Conditional on $\mathcal{F}_p(t)^{\complement}$, if $T^{(i)} > \bar{T}^{(i)}$ we have $\mathrm{UCB}_j^{(i)}(t) < \mathrm{LCB}_{j'}^{(i)}(t)$ for any $j, j' \in [K]$ with $\mu_j^{(i)} < \mu_{j'}^{(i)}$.*

Lemma 1.3 can similarly be written for arms. Following we extend the Lemma 5.3 from [1] to two-sided learning

Define the event $\mathcal{F}_a(t) = \left\{ \exists i \in [N],\ j \in [K] : |\hat{\eta}_i^{(j)} - \eta_i^{(j)}| > \sqrt{\frac{2\log t}{T_i^{(j)}}} \right\}$, an arm's bad event that some preference of the players are not estimated by the arms correctly at time t.

Lemma 1.4. *Conditional on $\cap_{t=1}^{T}(\mathcal{F}_p(t) \cup \mathcal{F}_a(t))^{\complement}$, after epoch $l_{\max}$, the player-optimal regret and arm-pessimal regret for the period $b^l T_0 - 2^l T_0 - K^2$ in epoch $l > l_{\max}$ is zero where*

$$l_{\max} = \min\left\{ l : \sum_{l'=1}^{l} 2^{l'} T_0 \geq 32K\log T/\Delta^2 \right\}. \tag{2}$$

Proof. Since in each exploration period all players propose to distinct arms using a round-robin fashion, no collision occurs and all players are accepted at each round of the exploration period in each epoch. Thus at the end of epoch $l_{\max}$ it holds that $T_j^{(i)} \geq 32\log T/(\Delta^{(i)})^2$ and $T_i^{(j)} \geq 32\log T/(\Delta'^{(j)})^2$ for any $i \in [N]$ and $j \in [K]$ since $\Delta = \min_{i\in[N], j\in[K]}\{\Delta^{(i)}, \Delta'^{(j)}\}$.
Now according to Lemma 1.3, when $T_j^{(i)} \geq 32\log T/(\Delta^{(i)})^2$ for any arm a_j, player p_i finds a permutation $\sigma^{(i)}$ over arms such that $\mathrm{LCB}_{\sigma_k^{(i)}}^{(i)} < \mathrm{UCB}_{\sigma_{k+1}^{(i)}}^{(i)}$ for any $k \in [K-1]$. Similarly, for any player p_i, arm a_j finds a permutation $\sigma'^{(j)}$ over players such that $\mathrm{LCB}_{\sigma_k^{(j)}}^{(j)} < \mathrm{UCB}_{\sigma_{k+1}^{(j)}}^{(j)}$ for any $k \in [N-1]$.
This implies that after $l > l_{\max}$, all players and arms finds a permutation of arms and players resp. with disjoint confidence intervals. Since, Gale-Shapley algorithm will last at most K^2 steps, we will have zero regret in the period $b^l T_0 - 2^l T_0 - K^2$ conditioned on the event $\cap_{t=1}^{T}(\mathcal{F}_p(t) \cup \mathcal{F}_a(t))^{\complement}$. □

1.1.1 Proof of Theorem 1

Proof of Theorem 1. From Lemma 1.3 after $t_{\exp} = 32K\log T/\Delta^2$, conditioned on the event $\cap_{t=1}^{T}(\mathcal{F}_p(t))^{\complement}$ every player p_i finds *true* preference ranking over arms and similarly, conditioned on the event $\cap_{t=1}^{T}(\mathcal{F}_p(t))^{\complement}$ every arm a_j finds correct preference ranking over players, thus the ith and $(N+j)^{\text{th}}$ entry of blackboard is 1. Thus everyone enters Gale-Shapley rounds after $t_{\exp}$

number of exploration rounds. Thus the optimal stable regret for player p_i is

$$\begin{aligned}
\overline{RP}_i(T) &= \mathbb{E}\left[\sum_{t=1}^{T} \mu^{(i)}_{\bar{m}_i} - X^{(i)}(t)\right] \\
&\leq \mathbb{E}\left[\sum_{t=1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i\}\Delta^{(i)}_{\max}\right] \\
&\leq N\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i, \mathcal{F}_p(t)^{\complement}\}\right]\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\mathcal{F}_p(t)\} + 1\{\mathcal{F}_a(t)\}\right]\Delta^{(i)}_{\max} \\
&\leq N\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i, \mathcal{F}_p(t)^{\complement}\}\right]\Delta^{(i)}_{\max} + \frac{2NK\pi^2}{3}\Delta^{(i)}_{\max} \\
&\leq N\Delta^{(i)}_{\max} + \frac{32K\log T}{\Delta^2}\Delta^{(i)}_{\max} + \frac{2NK\pi^2}{3}\Delta^{(i)}_{\max}
\end{aligned}$$

□

1.1.2 Proof of Theorem 2

We will find an equation of $l_{\max}$, from the definition we have and use the fact that $b > 2$ to get

$$\begin{aligned}
l_{\max} &= \left\lceil \log\left(\frac{32K\log T}{T_0(\Delta^{(i)})^2} + 2\right) - 1\right\rceil = \left\lceil \log\left(\frac{16K\log T}{T_0(\Delta^{(i)})^2} + 1\right)\right\rceil \\
\implies \log\left(\frac{16K\log T}{T_0(\Delta^{(i)})^2} + 1\right) &\leq l_{\max} < \log\left(\frac{16K\log T}{T_0(\Delta^{(i)})^2} + 1\right) + 1 = \log\left(\frac{32K\log T}{T_0(\Delta^{(i)})^2} + 2\right)
\end{aligned}$$

Define $\tilde{l}$, the epoch number required to complete T number of interaction rounds. We have from the definition

$$\begin{aligned}
\sum_{l=1}^{\tilde{l}} b^l T_0 &= T - N \\
\frac{b^{\tilde{l}+1} - b}{b-1} T_0 &= T - N \\
b^{\tilde{l}+1} &= \frac{(T-N)}{T_0}(b-1) + b \\
\tilde{l} &= \log_b\left(\frac{b-1}{b}\frac{(T-N)}{T_0} + 1\right) < \log\left(\frac{T-N}{T_0} + 1\right)^{\log_b 2} = \log\left(\frac{T-N}{T_0} + 1\right)^{\gamma}
\end{aligned}$$

Proof of Theorem 2. The optimal stable regret for player p_j is the sum of regret before and after $l_{\max}$.

$$\begin{aligned}
\overline{RP}_i(T) &= \mathbb{E}\left[\sum_{t=1}^{T} \mu^{(i)}_{\bar{m}_i} - X^{(i)}(t)\right] \\
&\leq \mathbb{E}\left[\sum_{t=1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i\}\Delta^{(i)}_{\max}\right] \\
&\leq N\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i, \mathcal{F}_p(t)^{\complement}\}\right]\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\mathcal{F}_p(t)\} + 1\{\mathcal{F}_a(t)\}\right]\Delta^{(i)}_{\max} \\
&\leq N\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i, \mathcal{F}_p(t)^{\complement}\}\right]\Delta^{(i)}_{\max} + \frac{2NK\pi^2}{3}\Delta^{(i)}_{\max}
\end{aligned}$$

Let $T_1 = \sum_{l=1}^{l_{\max}} b^l T_0$ and $T_2 = \sum_{l=l_{\max}+1}^{\tilde{l}} b^l T_0$ be the time-period before and after epoch $\tilde{l}$ with $T - N = T_1 + T_2$. Thus we have

$$R(T) = \mathbb{E}\left[\sum_{N+1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i, \mathcal{F}_p(t)^{\complement}\}\right] \Delta_{\max}^{(i)} = \mathbb{E}\left[\sum_{l=1}^{l_{\max}} b^l T_0\right] \Delta_{\max}^{(i)} + \mathbb{E}\left[\sum_{l=l_{\max}+1}^{\tilde{l}} 2^l T_0 + K^2\right] \Delta_{\max}^{(i)}$$

since the regret before epoch $l_{\max}$ i.e. for $l \leq l_{\max}$ is $b^l T_0$ and after $l_{\max}$ we have non-zero regret only from exploration and Gale-Shapley rounds due to Lemma 1.4. Thus the total regret is

$$\begin{aligned}
R(T) &= \mathbb{E}\left[\sum_{l=1}^{l_{\max}} b^l T_0\right] \Delta_{\max}^{(i)} + \mathbb{E}\left[\sum_{l=l_{\max}+1}^{\tilde{l}} 2^l T_0 + K^2\right] \Delta_{\max}^{(i)} \\
&\leq T_0 b^{l_{\max}+1} \Delta_{\max}^{(i)} + T_0(2^{\tilde{l}+1} - 2^{l_{\max}+1}) \Delta_{\max}^{(i)} + K^2 \tilde{l} \Delta_{\max}^{(i)} \\
&= 2^{\tilde{l}+1} T_0 \Delta_{\max}^{(i)} + (b^{l_{\max}+1} - 2^{l_{\max}+1}) T_0 \Delta_{\max}^{(i)} + K^2 \tilde{l} \Delta_{\max}^{(i)} \\
&\leq 2^{\tilde{l}+1} T_0 \Delta_{\max}^{(i)} + (l_{\max}+1) b^{l_{\max}} (b-2) T_0 \Delta_{\max}^{(i)} + K^2 \tilde{l} \Delta_{\max}^{(i)} && \text{(convexity of } x^{l_{\max}+1} \text{ for } l_{\max} \geq 1) \\
&\leq 2\left(\frac{T}{T_0}\right)^{\gamma} T_0 \Delta_{\max}^{(i)} + \log\left(\frac{64K \log T}{T_0 (\Delta^{(i)})^2} + 4\right)\left(\frac{32K \log T}{T_0 (\Delta^{(i)})^2} + 2\right)^{1/\gamma} (2^{1/\gamma} - 2) T_0 \Delta_{\max}^{(i)} + K^2 \gamma \log\left(\frac{T}{T_0}\right) \Delta_{\max}^{(i)}
\end{aligned}$$

We use the following inequality of convex functions

$$f(x) - f(y) \leq \nabla f(x)(x - y).$$

Thus we have in total

$$\begin{aligned}
\overline{RP}_i(T) \leq N \Delta_{\max}^{(i)} &+ 2\left(\frac{T}{T_0}\right)^{\gamma} T_0 \Delta_{\max}^{(i)} + \log\left(\frac{64K \log T}{T_0 (\Delta^{(i)})^2} + 4\right)\left(\frac{32K \log T}{T_0 (\Delta^{(i)})^2} + 2\right)^{1/\gamma} (2^{1/\gamma} - 2) T_0 \Delta_{\max}^{(i)} \\
&+ \ \gamma K^2 \log\left(\frac{T}{T_0}\right) \Delta_{\max}^{(i)} + \frac{2NK\pi^2}{3} \Delta_{\max}^{(i)}
\end{aligned}$$

For T_2 to be defined correctly we need that

$$\begin{aligned}
\log\left(\frac{T-N}{T_0} + 1\right)^{\gamma} > \tilde{l} \geq l_{\max} + 1 &\geq \log\left(\frac{32K \log T}{T_0 (\Delta^{(i)})^2} + 2\right) \\
\implies \left(\frac{T-N}{T_0}\right)^{\gamma} &\gtrsim \frac{32K \log T}{T_0 (\Delta^{(i)})^2} \\
T_0 &\gtrsim \left(\frac{32K \log T}{(\Delta^{(i)})^2 (T-N)^{\gamma}}\right)^{\frac{1}{1-\gamma}} \quad \forall i \\
\implies T_0 &\gtrsim \left(\frac{32K \log T}{\Delta^2 (T-N)^{\gamma}}\right)^{\frac{1}{1-\gamma}}
\end{aligned}$$

□

1.2 Arm's Learning

1.2.1 Known Gap

Algorithm 1: Expore-then-Gale-Shapley (view of arm a_j) with Known Gap Δ

Input : Exploration rounds $t_{\text{exp}} = \left\lceil \frac{32K \log T}{\Delta^2} \right\rceil$

`// Learning Preferences`

for $t = 1, \ldots, t_{\text{exp}}$ **do**
  Accept arm with estimated preferences
  Observe $Y^{(j)}_{A_j^{-1}(t)}(t)$ and update $\hat{\mu}^{(j)}_{A_j^{-1}(t)}, T^{(j)}_{A_j^{-1}(t)}$ if $\bar{A}_j^{-1}(t) = A_j^{-1}(t)$
end
Compute $\text{UCB}^{(j)}_n$ and $\text{LCB}^{(j)}_n$ for each $n \in [N]$
if $\exists \beta$ such that $\text{LCB}^{(j)}_{\beta_n} > \text{UCB}^{(j)}_{\beta_{n+1}}$ for any $n \in [N-1]$ **then**
  Preferences $= \beta$
else
  Preferences = arbitrary but fixed
end

`// Perform Gale-Shapley using the Preferences` $\beta^{(j)} = \beta = (\beta_1^{(j)}, \beta_2^{(j)}, \ldots, \beta_N^{(j)})$ `found`

Propose using β till acceptance
Initialize $s = 1$
for $t = t_{\text{exp}} + 1, \ldots, T$ **do**
  Accept player using the estimated preference $\beta^{(j)}$
  $s = s + 1$ if $\bar{A}_j^{-1}(t) == \emptyset$
end

1.2.2 `CA-ETC`

Algorithm 2: Base Algorithm: Expore-then-Gale-Shapley (view of arm a_j)

Input : Exploration rounds $2^l T_0$, Horizon $b^l T_0$

`// Learning Preferences`

for $t = \sum_{l'=1}^{l-1} 2^{l'} T_0 + 1, \ldots, \sum_{l'=1}^{l-1} 2^{l'} T_0 + 2^l T_0$ **do**
  Accept arm with estimated preferences
  Observe $Y^{(j)}_{A_j^{-1}(t)}(t)$ and update $\hat{\mu}^{(j)}_{A_j^{-1}(t)}, T^{(j)}_{A_j^{-1}(t)}$ if $\bar{A}_j^{-1}(t) = A_j^{-1}(t)$
end
Compute $\text{UCB}^{(j)}_n$ and $\text{LCB}^{(j)}_n$ for each $n \in [N]$
if $\exists \beta$ such that $\text{LCB}^{(j)}_{\beta_n} > \text{UCB}^{(j)}_{\beta_{n+1}}$ for any $n \in [N-1]$ **then**
  Preferences $= \beta$
else
  Preferences = arbitrary but fixed
end

`// Perform Gale-Shapley using the Preferences` $\beta^{(j)} = \beta = (\beta_1^{(j)}, \beta_2^{(j)}, \ldots, \beta_N^{(j)})$ `found`

Propose using β till acceptance
Initialize $s = 1$
for $t = 1, 2, \ldots, b^l T_0 - 2^l T_0$ **do**
  Accept player using the estimated preference $\beta^{(j)}$
  $s = s + 1$ if $\bar{A}_j^{-1}(t) == \emptyset$
end

Algorithm 3: Main Algorithm: Epoch-based `CA-ETC` (view of player a_j)

Input: Epoch length T_0, Parameter $\gamma \in (0,1)$ with $b = 2^{1/\gamma}$

for $l = 1, 2, \ldots$ **do**

  `Base Algorithm (Exploration rounds =` $2^l T_0$`, Horizon length =` $b^l T_0$`)`

end

1.2.3 Analysis

Theorem 1.5 (Regret of Algorithm 1)**.** *Suppose algorithm 1 is played for T iterations. Then, arm a_j incurs the regret of*

$$\underline{RA}_j(t) \leq \left(\frac{64K \log T}{\Delta^2} + K^2 + \frac{2NK\pi^2}{3} \right) \bar{\Delta}^{(j)}_{\max}.$$

Theorem 1.6 (Regret of Algorithm 3)**.** *Suppose `CA-ETC` is run with initial epoch length T_0 and input parameter $\gamma \in (0,1)$. Then, provided the initial epoch satisfies*

$$T_0 \gtrsim \left[\frac{32K \log T}{\Delta^2 (T-N)^\gamma} \right]^{\frac{1}{1-\gamma}},$$

the (arm-pessimal) regret for arm a_j is given by

$$\begin{aligned} \underline{RA}_j(t) \leq\ & T_0 \left(\frac{64K \log T}{T_0 (\Delta^{(j)})^2} + 4 \right)^{1/\gamma} \Delta^{(j)}_{\max} \\ & + 2T_0 \left(\frac{T-N}{T_0} + 1 \right)^{\gamma} \Delta^{(j)}_{\max} + K^2 \gamma \log \left(\frac{T-N}{T_0} + 1 \right) \Delta^{(j)}_{\max} + \frac{2NK\pi^2}{3} \Delta^{(j)}_{\max}. \end{aligned}$$

The proof of these follow the same steps as the proof for the players, and hence skipped.

1.3 Two-sided learning with (player, arm) broadcast

Assumption 1.7. All players observe the successfully accepted player for each arm at the end of the round.

Assumption 1.8. All arms observe the successfully accepted arm for each player at the end of the round.

We consider the above assumptions, which allow for each match being broadcasted to every agent. Below we present a simple extension of [1] for two-sided learning with these two assumption. We show that this incurs the same regret bound as the blackboard algorithm presented in the paper. Thus, our assumption of blackboard can be relaxed with these two assumptions.

1.3.1 Player's Learning

Algorithm 4: Expore-then-Gale-Shapley (view of player p_j)

```
// Perform index estimation
// Learning Preferences
for l = 1, 2, ... do
    F_l^p = False
    for t = Σ_{l'=1}^{l-1} 2^{l'} + 1, ..., Σ_{l'=1}^{l-1} 2^{l'} + 2^l do
        A_j(t) = a_{(Index+t-1)%K+1}  // Round-robin exploration
        Observe X^{(j)}_{A_j(t)}(t) and update μ̂^{(j)}_{A_j(t)}, T^{(j)}_{A_j(t)} if Ā_j(t) = A_j(t)
    end
    Compute UCB_k^{(j)} and LCB_k^{(j)} for each k ∈ [K]
    if ∃σ such that LCB^{(j)}_{σ_k} > UCB^{(j)}_{σ_{k+1}} for any k ∈ [K − 1] then
        F_l^p = True and preferences = σ
    end
    Initialize O_l = ∅
    A_j(t) = a_Index if F_l^p == True and A_j(t) = ∅ otherwise
    Update O_l = ∪_{j'∈[N]} {Ā_{j'}(t)}  // set of successfully accepted arms for all players
    if |O_l| == N then
        Perform Gale-Shapley using the Preferences σ^{(j)} = σ = (σ_1^{(j)}, σ_2^{(j)}, ..., σ_K^{(j)}) found; t_2 = t
    end
end
// Gale-Shapley
Propose using σ till acceptance
Initialize s = 1
for t = t_2, t_2 + 1, ..., T_0 do
    A_j(t) = a_{σ_s^{(j)}}
    s = s + 1 if Ā_j(t) == ∅
end
```

1.3.2 Arm's Learning

Algorithm 5: Base Algorithm: Expore-then-Gale-Shapley (view of arm a_j)

```
// Learning Preferences
for l = 1, 2, ... do
    F_l^a = False
    for t = Σ_{l'=1}^{l-1} 2^{l'} + 1, ..., Σ_{l'=1}^{l-1} 2^{l'} + 2^l do
        Observe Y^{(j)}_{A_j^{-1}(t)}(t) and update μ̂^{(j)}_{A_j^{-1}(t)}, T^{(j)}_{A_j^{-1}(t)} if Ā_j^{-1}(t) = A_j^{-1}(t)
    end
    Compute UCB_n^{(j)} and LCB_n^{(j)} for each n ∈ [N]
    if ∃β such that LCB^{(j)}_{σ_n} > UCB^{(j)}_{σ_{n+1}} for any n ∈ [N − 1] then
        F_l^a = True and preferences = σ
    end
    Initialize O_l = ∅
    Accept the proposed arm if F_l^a == True and reject otherwise
    Update O_l = ∪_{j'∈[N]} {Ā_{j'}^{-1}(t)} // set of successfully accepted players for all arms
    if |O_l| == N then
        Perform Gale-Shapley using the Preferences σ^{(j)} = σ = (σ_1^{(j)}, σ_2^{(j)}, ..., σ_K^{(j)}) found; t = t_2
    end
end
// Gale-Shapley
Propose using σ till acceptance
Initialize s = 1
for t = t_2, t_2 + 1, ..., T do
    A_j(t) = a_{σ_s^{(j)}}
    s = s + 1 if Ā_j(t) == ∅
end
```

The algorithm proceeds in three phases

- **Phase 1**:
  Player p_i's POV: Propose arm a_1 till it gets matched and denote Index as the time at which it gets matched.
  Result: Since at the start of the algorithm arm a_1 will have arbitrary preferences over player's, for each player Index will be the preference of arm a_1 over the players. Note that Index will be distinct for each player.

- **Phase 2**:
  This phase contains sub-phases of increasing duration $2^l + 1$ for sub-phase l.
  Exploration: In each sub-phase l of duration 2^l, let each player p_i to propose a distinct arm in round-robin fashion using Index. Based on the interaction, player p_i and arm a_j update their $\text{UCB}_j^{(i)}, \text{LCB}_j^{(i)}$ and $\text{UCB}_i^{(j)}, \text{LCB}_i^{(j)}$ resp.
  Monitoring: Here, we let player p_i play arm a_{Index}, if p_i has estimated an accurate preference ranking over arms. This happens when, for p_i the confidence interval for each arm a_j $[\text{LCB}_j^{(i)}, \text{UCB}_j^{(i)}]$ does not overlap. [1]
  After this, player p_i (arm a_j) has to check whether all players and arms has also estimated correct preference ranking. If yes, we enter Phase 3 else we continue repeating sub-phases in Phase 2.

- **Phase 3**:
  Here the scenario is that every player and arm has estimated correct preference ranking. Thus, at this phase, each player and arm will run a Gale-Shapley algorithm to find an optimal stable match.

Now we will see how exactly the monitoring phase proceeds in Phase 2.

Q. How does player p_i know that all players and arms have estimated correct preference ranking?
Ans. In each sub-phase l, players (arms) will maintain their own flag F_l^p (F_l^a), which is set to true if they have estimated

[1] Similarly, arm a_j has estimated correct preference ranking if for each player p_i $[\text{LCB}_i^{(j)}, \text{UCB}_i^{(j)}]$ does not overlap.

corrected preference ranking. Note, since the algorithm is decentralized player p_i (arm a_j) does not know the flag of other players (arms).
Once F_l^p is true, i.e. player p_i has found correct preference ranking, it proposes arm a_{Index}. If arm a_{Index} has found the correct preference ranking as well i.e. F_l^a is true, then we let a_{Index} to accept player p_i, if not, arm a_{Index} should not accept player p_i.
Due to the assumption 1.7, each player forms a set $\mathcal{P}_l = \cup_{i' \in [N]} \{\bar{A}'_i\}$ which is set of successfully matched arms for all players. If $|\mathcal{P}_l|$ is N then it implies that all players and arms have estimated correct preference ranking.

Q. How does arm a_j know that all players and arms have estimated correct preference ranking?
Ans. Once player p_i estimates correct preference ranking, it proposes arm a_{Index}, and if arm a_{Index} has also estimated correct preference ranking, it accepts the proposal. Due to the assumption 1.8, each arm will thus form a set $\mathcal{A}_l = \cup_{i' \in [N]} \{\bar{A}'_i\}$ which is set of successfully matched arms for all players. If $|\mathcal{A}_l|$ is N then it implies that all players and arms have estimated correct preference ranking.

Recall that, $N \leq K$ was assumed hence we check the cardinality of set equal to N.

1.3.3 Analysis

Theorem 1.9 (Regret of Algorithm 4)**.** *Suppose algorithm 4 is played for T iterations. Then, player p_i incurs the player-optimal regret of*

$$\overline{RP}_i(t) \leq \left(N + \frac{64K \log T}{\Delta^2} + K^2 + \frac{2NK\pi^2}{3} \right) \Delta_{\max}^{(i)}.$$

Theorem 1.10 (Regret of Algorithm 5)**.** *Suppose algorithm 5 is played for T iterations. Then, arm a_j incurs the player-pessimal regret of*

$$\underline{RA}_j(t) \leq \left(\frac{64K \log T}{\Delta^2} + K^2 + \frac{2NK\pi^2}{3} \right) \bar{\Delta}_{\max}^{(j)}.$$

The proof of both the theorems are omitted as they exactly follow [1].

References

[1] F. Kong and S. Li, *Player-optimal Stable Regret for Bandit Learning in Matching Markets*, pp. 1512–1522. [Online]. Available: https://epubs.siam.org/doi/abs/10.1137/1.9781611977554.ch55